

Quantum Mechanics II Excercise 2

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1 Analytical problems

1.1 Ising model with nearest + next nearest interaction

The hamiltonian for the Ising model with nearest + next nearest interaction is

$$H = -J_1 \sum_i \sigma_i \sigma_{i+1} - J_2 \sum_i \sigma_i \sigma_{i+2} - B \sum_i \sigma_i \quad (1)$$

Since the boundries are negligible in the thermodynamic limit, we can select periodic boundary conditions $N+1 = 1, N+2 = 2$. The partition function is then

$$\begin{aligned} Z_N &= \sum_{\sigma_1=\pm 1} \cdots \sum_{\sigma_N=\pm 1} \exp \left[\beta J_1 \sum_i \sigma_i \sigma_{i+1} + \beta J_2 \sum_i \sigma_i \sigma_{i+2} + \beta B \sum_i \sigma_i \right] = \\ &= \sum_{\sigma_1=\pm 1} \cdots \sum_{\sigma_N=\pm 1} \prod_i \exp (\beta J_1 \sigma_i \sigma_{i+1} + \beta J_2 \sigma_i \sigma_{i+2} + \beta B \sigma_i) \end{aligned} \quad (2)$$

To create a transfer matrix, let us define a superspin $|\bar{\sigma}_i\rangle \in \mathbb{R}^4$ out of every two consecutive spins:

$$|\uparrow\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |\uparrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (3)$$

Which means the exponent in equation 2 is

$$\exp (\beta J_1 \sigma_i \sigma_{i+1} + \beta J_2 \sigma_i \sigma_{i+2} + \beta B \sigma_i) = \langle \bar{\sigma}_i | M | \bar{\sigma}_{i+1} \rangle \quad (4)$$

Where M is the transfer matrix:

$$M = \exp \left[\beta \begin{pmatrix} J_1 + J_2 + B & J_1 - J_2 + B & & \\ -J_1 + J_2 + B & -J_1 - J_2 + B & & \\ & & -J_1 - J_2 - B & -J_1 + J_2 - B \\ & & J_1 - J_2 - B & J_1 + J_2 - B \end{pmatrix} \right] \quad (5)$$

Naming the four vectors from equation 3 $b_1, \dots, b_4$, the partition function is

$$Z_N = \sum_{\bar{\sigma}_1=b_1 \dots b_4} \cdots \sum_{\bar{\sigma}_N=b_1 \dots b_4} \langle \bar{\sigma}_1 | M | \bar{\sigma}_2 \rangle \langle \bar{\sigma}_2 | M | \bar{\sigma}_3 \rangle \cdots \langle \bar{\sigma}_N | M | \bar{\sigma}_1 \rangle = \text{Tr}(M^{N/4}) \quad (6)$$

Where in the last equation we used the fact that M is translationally invariant. Since the trace is base independent, we can express it as the sum of the matrix's eigenvalues, and in the thermodynamic limit, all eigenvalues except the largest one will be negligible, and the partition function is

$$\lim_{N \rightarrow \infty} Z_N = \lambda_{\max}^N \quad (7)$$

This result makes sense because it gives us extensive free energy.

1.2 Ising model as a Lattice Gas

Given seven sites and three particles that could be in those sites, there is a $\binom{7}{3}$ options for their configuration, but we can make the situation much simpler. The first thing to notice is the translation invariance, so we can assume the first particle is in the first position, and simply multiply the options by 7 at the end. This should leave us with $\binom{7}{3}/7 = 5$ options, which is a very manageable number. Now let's look at the positions of the other two particles. We don't want to overcount by saying that placing the particles in positions (1, 6, 7) is distinct from placing them in (1, 2, 3). Those two situations are the same under translations, which we already accounted for. We can find duplicates by enumeration and remove them. The positions that are left after removing all duplicates are

$$b_1 = (1, 2, 3) \quad b_2 = (1, 2, 4) \quad b_3 = (1, 2, 5) \quad b_4 = (1, 2, 6) \quad b_5 = (1, 3, 5) \quad (8)$$

Using the Ising hamiltonian for each configuration, the magnetic field term gives $-3B$ for every configuration, while the interaction term is positive if the two sites are both occupied or both unoccupied. i.e. we start from $-7J$ and gain $2J$ for every wall:

$$Z_1 = e^{3\beta J + 3\beta B} \quad Z_2 = Z_3 = Z_4 = e^{-\beta J + 3\beta B} \quad Z_5 = e^{-5\beta J + 3\beta B} \quad (9)$$

In other words we have one situations with two walls, three situations with four walls and one situation with six walls. Summing over each of them and multiplying by seven for the translation invariance, we get:

$$\begin{aligned} Z &= e^{3\beta J + 3\beta B} + 3e^{-\beta J + 3\beta B} + e^{-5\beta J + 3\beta B} = \\ &= e^{3\beta B} (e^{3\beta J} + 3e^{-\beta J} + e^{-5\beta J}) \end{aligned} \quad (10)$$

2 Numeric Problems

2.1 Easy Numeric

2.1.1 The free energy and the partition function

The Helmholtz free energy is defined using the Legendere transform over the energy

$$F(T, V, N) = U(S, V, N) - TS \quad (11)$$

We can calculate the expected entropy using its definition

$$S = -k_B \ln(\Omega) \quad (12)$$

Where Ω is the multiplicity of the macrostate. We can define all macrostates of the system i where $p_i = \Omega_i/\Omega$ is the probability of the macrostate and get

$$\langle S \rangle = -k_B \sum_i p_i \ln(p_i) \quad (13)$$

Now, using the Boltzman factor as the probability, we get

$$\begin{aligned} \langle S \rangle &= -k_B \sum_i p_i \ln\left(\frac{e^{-\beta E_i}}{Z}\right) = -k_B \sum_i p_i [-\beta E_i - \ln Z] = \\ &= -k_B \left[-\beta \sum_i p_i E_i - \ln Z \sum_i p_i \right] = k_B \beta \langle E \rangle + k_B \ln Z \\ T \langle S \rangle &= \langle E \rangle + k_B T \ln Z \end{aligned} \quad (14)$$

Comparing this to equation (11) we get

$$F = -k_B T \ln Z \quad (15)$$

2.1.2 The Helmholtz free energy for the 1D Ising model

In this section we ran a simulation of the 1D Ising model, with $N \in [6, 12]$ and $\beta \in [0.1, 10]$ with jumps ever 0.1. We ran it in the ferromagnetic case $J = 1$ and in the anti-ferromagnetic case $J = -1$. For each simulation we calculated the free energy using the eigenvalue approximation and directly from the microstates of the system. We then plotted the difference between the approximation and the real value.¹

Let us first look at what we expect to see. For $B = 0$, the two eigenvalues of the transfer matrix of the 1-D Ising models become

$$\lambda_{\pm} = e^{\beta J} \cosh(\beta B) \pm \sqrt{e^{2\beta J} \cosh^2(\beta B) - 2 \sinh(2\beta J)} = e^{\beta J} \pm e^{-\beta J} = 2 \cosh(\beta J), 2 \sinh(\beta J) \quad (16)$$

the exact theoretic solution for the free energy of the 1-D Ising model is given by

$$F_{theo,exact} = -\beta^{-1} \ln(\lambda_+^N + \lambda_-^N) = -\beta^{-1} N \ln \left(\sqrt[2]{(2 \cosh(\beta J))^N + (2 \sinh(\beta J))^N} \right) \quad (17)$$

While the theoretic value free energy as taken from the largest eigenvalue is (for $B = 0$):

$$F_{theo,approx} = -\beta^{-1} \ln(\lambda_+^N) = -\beta^{-1} N \ln(2 \cosh(\beta J)) \quad (18)$$

The difference between them is

$$\Delta F = -\beta^{-1} \ln(1 + \tanh(\beta J)^N) \quad (19)$$

Which has two minima for even N s, symmetric around $\beta J = 0$, and a single minimum for odd N s. This means we expect to find a maximum error in both the ferromagnetic case and for even N s in the antiferromagnetic case.

The results can be seen in figures figure 1 and figure 2.

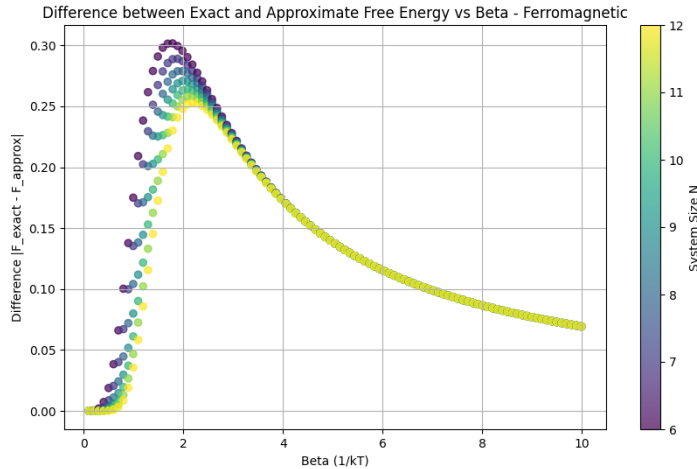


Figure 1: The difference in the Helmholtz free energy between the exact value and approximation value as a function of β for the ferromagnetic case ($J = 1$) for different system sizes N .

First thing we can see that in the ferromagnetic case the higher the number of particles, the more accurate the approximation is. In contrast, in the antiferromagnetic case, this is true only for odd and even N s separately.

¹The code for this section can be found in https://github.com/idanstark42/masters-homework/blob/master/sm1/ex1/ising_1d.py

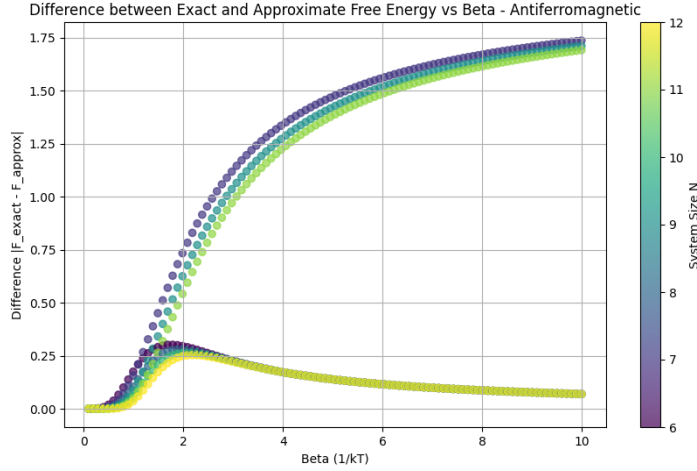


Figure 2: The difference in the Helmholtz free energy between the exact value and approximation value as a function of β for the anti-ferromagnetic case ($J = -1$) for different system sizes N .

In general, below a certain temperature, the error of odd numbered models grows with β . This is exactly what we thought would happen. We can understand this result physically as well: The antiferromagnetic case with an odd number of particles and periodic boundary conditions, the ground state has degeneracy N (two neighboring sites must be with the same spin, and there are N options to which two sites it would be). This means that ergodicity is broken behind the two cases of the Ising model.

The second thing we notice is that for the ferromagnetic case and the even- N antiferromagnetic case, there is a maximum error, exactly as expected. Finally, to see where the point where the approximation fails, we can zoom into the range $\beta \in [0, 1]$. This can be found in figures 3 and 4. It can be seen that the more particles the system has, the more accurate the approximation is, but it always starts to fail around $\beta J = 1/2$

2.2 1D model with frustration

Let the hamiltonian of a triangular ladder be

$$H = -J \sum_i \sigma_i^L \sigma_{i+1}^L + \sigma_i^U \sigma_{i+1}^U + \sigma_i^L \sigma_i^U + \sigma_i^U \sigma_{i+1}^L \quad (20)$$

Let $\tilde{\sigma}_i$ be the superspin in $\mathbb{R}^4$ that encodes the information about σ_i^L and σ_i^U :

$$\begin{aligned} \tilde{\sigma}_i = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} &\rightarrow \sigma_i^L = 1, \sigma_i^U = 1 & \tilde{\sigma}_i = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} &\rightarrow \sigma_i^L = -1, \sigma_i^U = 1 \\ \tilde{\sigma}_i = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} &\rightarrow \sigma_i^L = 1, \sigma_i^U = -1 & \tilde{\sigma}_i = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} &\rightarrow \sigma_i^L = -1, \sigma_i^U = -1 \end{aligned} \quad (21)$$

The then partition function can be written as

$$Z = \sum_{\tilde{\sigma}_1} \cdots \sum_{\tilde{\sigma}_N} \prod_i \langle \tilde{\sigma}_i | M | \tilde{\sigma}_{i+1} \rangle \quad (22)$$

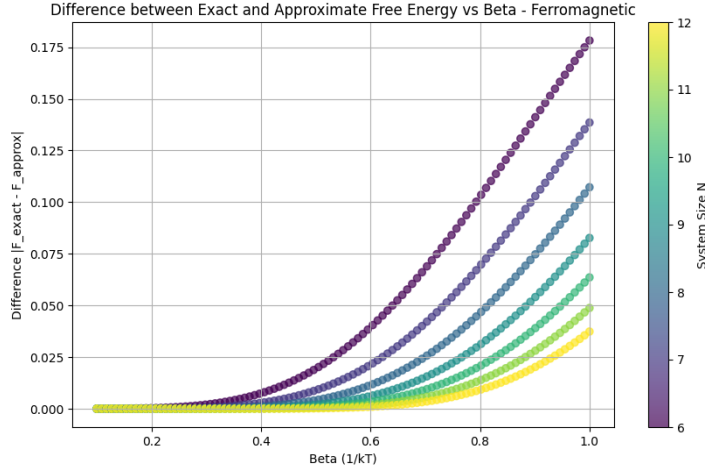


Figure 3: The difference in the Helmholtz free energy between the exact value and approximation value as a function of β for the ferromagnetic case ($J = 1$) for different system sizes N . Zoomed in to $\beta \in [0.1, 1.0]$

Where M is the transition matrix

$$M = \begin{pmatrix} e^{-4\beta J} & 1 & e^{-2\beta J} & e^{2\beta J} \\ 1 & 1 & e^{2\beta J} & e^{2\beta J} \\ e^{2\beta J} & e^{2\beta J} & 1 & 1 \\ e^{2\beta J} & e^{-2\beta J} & 1 & e^{-4\beta J} \end{pmatrix} \quad (23)$$

This makes the particle function

$$Z = \text{Tr}(M^N) = \sum_i \lambda_i^N \quad (24)$$

And in the thermodynamic limit, only the highest eigenvalue remains. This changes based on whether J is positive or negative. The eigen values are (from highest to lowest)

$$\begin{aligned} \lambda_1 &= \frac{1}{2} \left(2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)} \right) \\ \lambda_2 &= \frac{1}{2} \left(-2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 - e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)} \right) \\ \lambda_3 &= \frac{1}{2} \left(-2e^{2\beta J} + 1 + e^{-4\beta J} - e^{-4\beta J} \sqrt{(1 - e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)} \right) \\ \lambda_4 &= \frac{1}{2} \left(2e^{2\beta J} + 1 + e^{-4\beta J} - e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)} \right) \end{aligned} \quad (25)$$

We can see them in figure 6: For the ferromagnetic case, $J = 1$, the two middle eigenvalues λ_2, λ_3 are not defined in the region $\beta > 0$, because of the negative value in the square root. But, in both cases $\lambda_{max} = \lambda_1$:

$$F/N = \frac{1}{2} \left(2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)} \right) \quad (26)$$

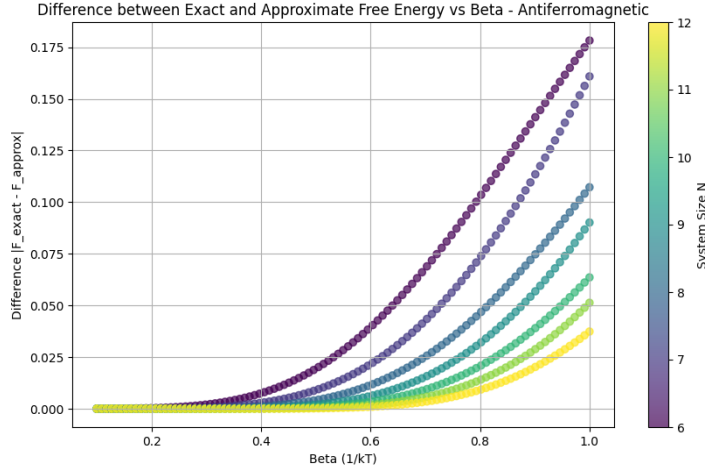


Figure 4: The difference in the Helmholtz free energy between the exact value and approximation value as a function of β for the anti-ferromagnetic case ($J = -1$) for different system sizes N . Zoomed in to $\beta \in [0.1, 1.0]$

Now let us find the correlation function $\langle \sigma_1^L \sigma_{1+r}^L \rangle$. Without loss of generality, let $i = 1$. The correlation function is, by definition

$$\langle \sigma_1^L \sigma_{1+r}^L \rangle = \frac{1}{Z} \sum_{\tilde{\sigma}_1} \cdots \sum_{\tilde{\sigma}_N} \langle \tilde{\sigma}_1 | AM | \tilde{\sigma}_2 \rangle \cdots \langle \tilde{\sigma}_r | AM | \tilde{\sigma}_{r+1} \rangle \cdots \langle \tilde{\sigma}_N | M | \tilde{\sigma}_{N+1} \rangle \quad (27)$$

Where

$$A = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \quad (28)$$

Thus

$$\langle \sigma_1^L \sigma_{1+r}^L \rangle = \frac{1}{Z} \text{Tr}(AM^{r-1}AM^{N-r+1}) \quad (29)$$

To perform the power over the matrix M , we need to diagonalize it $M = C^{-1}DC$:

$$\langle \sigma_1^L \sigma_{1+r}^L \rangle = \frac{1}{Z} \text{Tr}(AC^{-1}D^{r-1}CAC^{-1}M^{N-r+1}C) = \frac{1}{Z} \text{Tr}(CAC^{-1}D^{r-1}CAC^{-1}M^{N-r+1}) = \frac{1}{Z} \text{Tr}(BD^{r-1}BD^{N-r+1}) \quad (30)$$

Where $B = CAC^{-1}$. Writing M using its eigenvalues λ_i , the matrix multiplication becomes a linear combination of products of two eigenvalues in each term $\lambda_i^{r-1} \lambda_j^{N-r+1}$. Assigning $Z = \lambda_1^N$, every term that doesn't have λ_1^{N-r+1} is negligible, and we are left with the linear combinations of the terms that include it:

$$\langle \sigma_1^L \sigma_{1+r}^L \rangle = a_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^{r-1} + a_3 \left(\frac{\lambda_3}{\lambda_1} \right)^{r-1} + a_4 \left(\frac{\lambda_4}{\lambda_1} \right)^{r-1} \quad (31)$$

Since in our superspin model this is a 1-D Ising model with closest neighbor interaction, there is no phase transition and the correlation should decay as $r \rightarrow \infty$. Therefore $a_1 = 0$. Moreover, we can neglect the smaller ratios and get the well known relation

$$\langle \sigma_1^L \sigma_{1+r}^L \rangle = e^{-r/\xi} \quad \xi = \frac{1}{\ln \left(\frac{\lambda_{\max}}{\lambda_{\text{second}}} \right)} \quad (32)$$

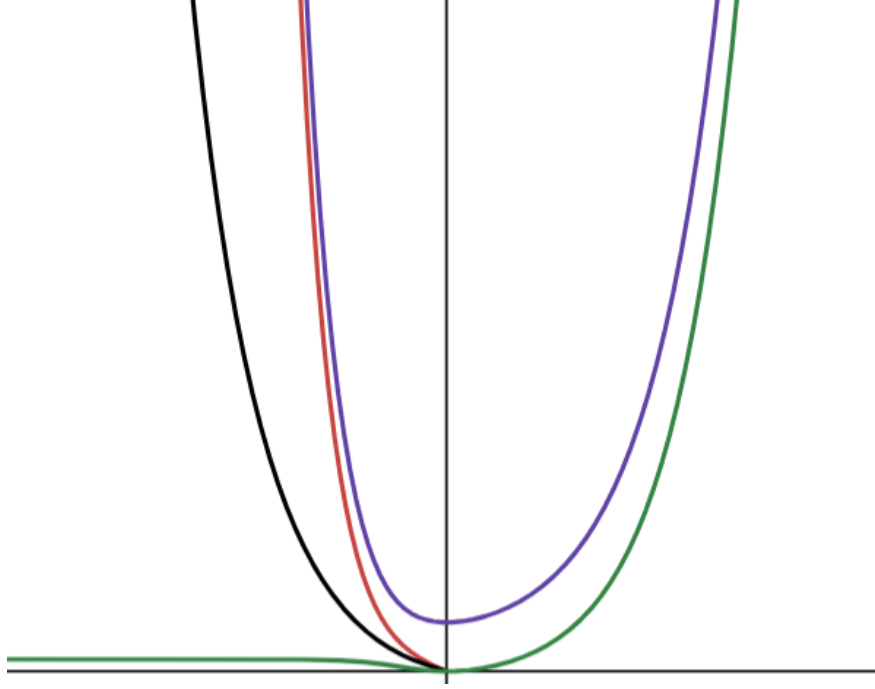


Figure 5: The eigenvalues of the transfer matrix M as a function of $J\beta$. Since $\beta > 0$, the right half of the plane shows the ferromagnetic case while the left half of the plane shows the anti-ferromagnetic for both ferromagnetic ($J = 1$) and anti-ferromagnetic ($J = -1$) cases.
purple = λ_1 , red = λ_2 , black = λ_3 , green = λ_4

For the ferromagnetic case $J = 1$, we remember that λ_2 is not defined, and therefore $\lambda_{second} = \lambda_4$

$$\xi = \left[\ln \left(\frac{2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)}}{2e^{2\beta J} + 1 + e^{-4\beta J} - e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)}} \right) \right]^{-1} \quad (33)$$

And for the anti-ferromagnetic case $J = -1$:

$$\xi = \left[\ln \left(\frac{2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 + e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)}}{-2e^{2\beta J} + 1 + e^{-4\beta J} + e^{-4\beta J} \sqrt{(1 - e^{2\beta J})^3 (4e^{4\beta J} - 3e^{\beta J} + 1)}} \right) \right]^{-1} \quad (34)$$

Those correlation distances can be seen as a function of β in figure 6: Unsurprisingly, in both cases, the correlation length is zero at the hot limit ($\beta = 0$) and diverges at the cold limit ($\beta = \infty$).

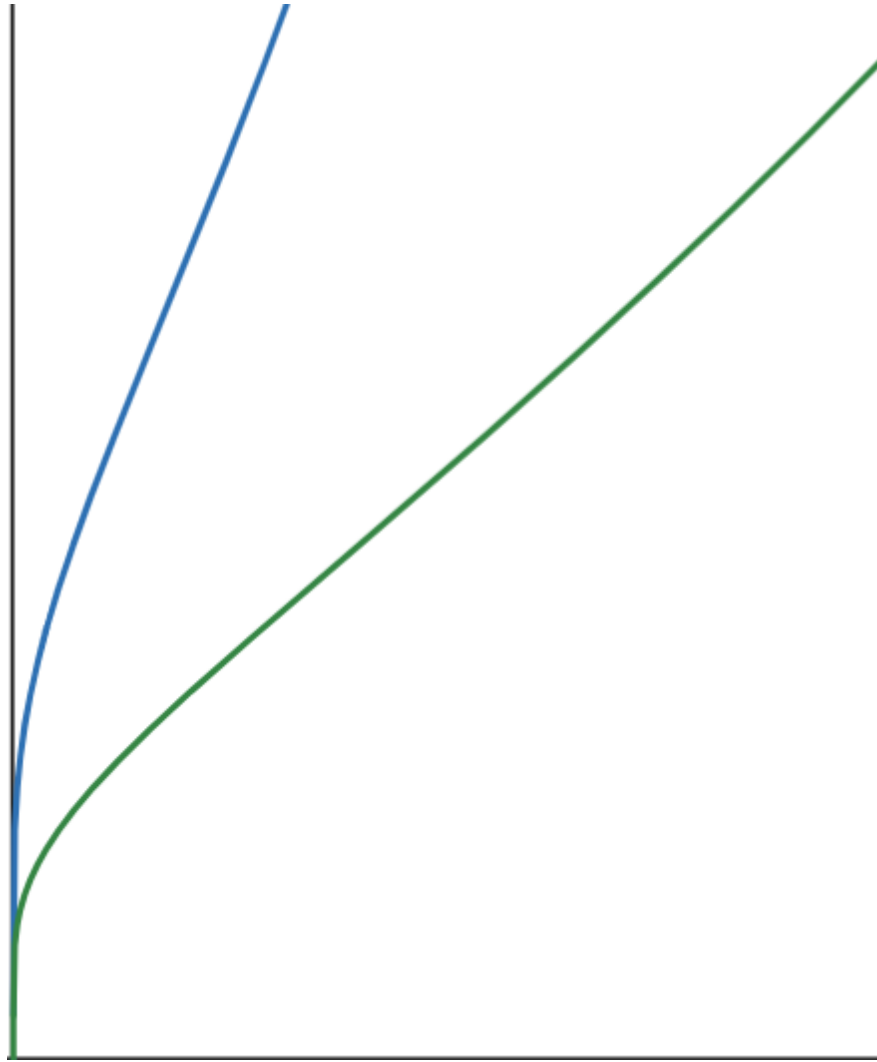


Figure 6: The correlation distance ξ as a function of β for both ferromagnetic ($J = 1$) and anti-ferromagnetic ($J = -1$) cases.
green = ferromagnetic, blue = anti-ferromagnetic